JOHN PUMAYALLI

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EDUCATION

Stanford University **Stanford, CA**
M.S. in Computer Science (Coterminal), Specialization: ***Artificial Intelligence*** *Expected March 2028*
B.S. in Symbolic Systems (Concentration: Artificial Intelligence) | GPA: 3.61/4.00 *Expected June 2027*
Relevant Coursework: Machine Learning (CS 229), Artificial Intelligence (CS 221), Decision Making under Uncertainty (CS 238), Design and Analysis of Algorithms (CS 161), Probability for Computer Scientists (CS 109)

Telluride High School, Telluride, CO – 4.34 GPA, Magna Cum Laude, Reagan Leadership Medal *June 2023*

ENGINEERING AND RESEARCH EXPERIENCE

Stanford Graduate School of Business, Macro Organizational Behavior **Stanford, CA**
Software Engineer and Research Assistant to Prof. Julien Clement (OrgOS) *March 2026–Present*
- Build and run the platform for OrgOS, an LLM pipeline that maps a company's teams, workstreams, and decisions from its Slack messages, on Google Cloud (Cloud Run, Postgres/pgvector, KMS) with Terraform
- Shipped per-pull-request preview environments (each PR runs its own backend, AI agent, and frontend on an isolated, sanitized database clone); closed security-audit gaps so only admins can reach customer data
- Found ~$700/month (~48%) in cloud savings by right-sizing services to p99 utilization; traced dashboard timeouts to stale Postgres statistics and cut the query from 7.5 s to 43 ms

Stanford University, Department of Communication **Stanford, CA**
Research Paper, "Talking to the Machine Teammate" (COMM 324, Prof. Jeff Hancock) *September 2026–Present*
- Analyzed 7.7M agent-authored PRs and 2,800+ repos in Python to study how reviewers treat AI agents
- Found agent-authored PRs merge 6.4 points less often than human PRs ($p = .008$), mostly from bot-account agents

Flourish, *Front-End Developer* – ML health-tracking app (React Native, Node.js, MongoDB) *Summer 2022*
Colorado School of Mines, *Research Assistant to Dr. Chuan Yue* – browser-extension privacy *Summer 2021*

MACHINE LEARNING PROJECTS

The Puma: Catan 1v1 Bot Trained from Scratch by Self-Play *November 2026*
- Trained a 279K-parameter graph neural network with PPO in PyTorch on 4.0M self-play games (779M positions) in 46 hours on a home PC, with 11 CPU workers feeding batched GPU inference
- Added AlphaZero-style tree search at play time; wins 195–200 of 200 games vs. Catanatron's AlphaBeta bot and 60% vs. a 1462-rated OpenHex ladder bot

Beyond Pip Counting: Interpretable ML for Catan Placement *CS 229 Final Project*
- Engineered 52 features (resource scarcity, turn order, opponent competition) over ~100K simulated games; lifted win-prediction AUC from 0.54 (pip-count baseline) to 0.73 with XGBoost
- Explained predictions with SHAP and distilled the model into a shallow decision tree whose 8 human-readable rules recover 62% of its edge over the baseline

Colonist.io 1v1 Post-Game Analyzer *catan-bot-two.vercel.app*
- Chrome extension records live games over WebSocket; Monte Carlo rollouts score every move and flag blunders

TEACHING AND LEADERSHIP

Stanford University, Department of Computer Science **Stanford, CA (Online)**
Section Leader, Probability for AI (Prof. Chris Piech) *October 2026–Present*
- Lead a weekly live section of a free Stanford course teaching the math behind modern AI to learners worldwide

Alumni Chair, Sigma Phi Epsilon – formerly Social Chair, Head of Member Development *September 2024–Present*
Counselor, Stanford Sierra Camp – led daily programming for 300+ guests per week *Summer 2025*

CULTURAL AND COMMUNITY INVOLVEMENT

Mestiza Qoyllacha, *Member and Performer* – Indigenous dance collective, Sacred Valley, Peru *2012–Present*

SKILLS AND ADDITIONAL INFORMATION

Programming: Python, SQL, JavaScript, C++, Java, Swift, Bash
Machine Learning: PyTorch, scikit-learn, XGBoost, SHAP, pandas, NumPy, LLM APIs (Gemini, Claude, Vertex AI)
Infrastructure: Google Cloud (Cloud Run, Cloud SQL, KMS, IAM), Terraform, GitHub Actions, Docker, PostgreSQL
Languages: Spanish (native), English (native), Italian (intermediate) | Interests: Tournament bridge, surfing